Software Test Plan (STP) - Online Exam Portal

Project: Online Exam Portal

Version: 1.0

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Status: Draft

1. Introduction

Purpose: This document defines the test plan for the Online Exam Portal v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers student registration and authentication, exam selection, question display, answer submission, exam timer, automatic evaluation, result generation, and administrator functions. Third-party services and infrastructure outside the project scope are excluded.

References: Online Exam Portal SRS v1.0, Design Specifications v1.0, and project requirements.

Definitions: OEP (Online Exam Portal), SRS (Software Requirements Specification), UAT (User Acceptance Testing), RTM (Requirements Traceability Matrix).

2. Test Items

- Student Registration and Authentication module
- Exam Selection module
- Question and Answer module
- Exam Timer module
- Exam Submission module
- Automatic Evaluation and Result module
- Admin/Teacher interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

- OEP-F-001: Validate student login
- OEP-F-002: Validate student registration
- OEP-F-003: Display available exams
- OEP-F-004: Display questions and answer options
- OEP-F-005: Maintain exam countdown timer
- OEP-F-006: Save selected answers
- OEP-F-007: Submit examination
- OEP-F-008: Automatically evaluate submitted answers
- OEP-F-009: Generate and display examination results
- OEP-F-010: Admin creates and manages exams and questions
- OEP-NF-001: Acceptable response time for portal operations

- OEP-NF-002: Support multiple users accessing an examination

4. Features Not to be Tested

- Third-party payment gateway integrations
- External email/SMS services not implemented by the project
- Hardware maintenance and device firmware
- Internal infrastructure of third-party hosting providers

5. Test Approach / Strategy

Levels:

- Unit tests (module-level)
- Integration tests (student portal ↔ database/backend)
- System tests (end-to-end Online Exam Portal functionality)
- Acceptance tests (UAT)

Types:

- Functional testing (core features)
- Regression testing
- Performance testing (response time and concurrent users)
- Usability testing (UI clarity and accessibility)

Entry Criteria: Stable build delivered, test data available, and test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, and all acceptance criteria satisfied.

5.1 Security Validation

- Validate secure authentication and password handling
- Verify unauthorized users cannot access restricted exams or administration functions
- Validate input fields against invalid and unexpected input
- Verify session management and logout behavior
- Verify sensitive information is not unnecessarily exposed

6. Test Environment

Hardware: Laptop/Desktop computer with stable internet connection.

Software: Online Exam Portal v1.0, supported web browser, application server, and database.

Tools: Selenium (UI automation), Postman (API testing, if applicable), JMeter (performance testing, if applicable), and Jira or equivalent defect-tracking tool.

Test Data: Dummy student accounts, administrator accounts, sample exams, questions, answer options, correct answers, and sample results.

7. Test Schedule

Milestones:

- Test case design: 18-Sep-2026
- Environment setup: 20-Sep-2026
- Test execution start: 21-Sep-2026
- Test execution end: 30-Sep-2026
- UAT: 01-Oct-2026 to 03-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Arvind Kothamangala	Prepare plan and coordinate execution
Test Engineer	Akasha	Design and execute test cases, log defects
Developer	Yashas K C	Support defect fixes and triage
Project Owner/Guide	B Anusha	Approve test results and sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early test builds from the development team
Test environment downtime	Maintain a backup test environment
High number of users during an exam	Perform performance/load testing before deployment
Incorrect result calculation	Use multiple positive and negative evaluation test cases
Timer malfunction	Test timer under normal, boundary, refresh, and submission conditions

11. Assumptions & Dependencies

- The application server and database will be available during testing.
- Test student and administrator accounts will be provided before execution.
- Sample exams, questions, and correct answers will be available.
- Required browsers and testing tools will be available.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment is unavailable for more than 4 hours
- Build is too unstable and blocks more than 30% of planned test cases
- Database or application server failure prevents meaningful testing

Resume testing if:

- Blocking defects are resolved
- Environment is stabilized
- A stable build is available for testing

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.

Example:

- OEP-F-001 (Student login) → TC-Login-01, TC-Login-02
- OEP-F-005 (Exam timer) → TC-Timer-01, TC-Timer-02
- OEP-F-008 (Automatic evaluation) → TC-Eval-01, TC-Eval-02
- OEP-NF-001 (Response time) → TC-Perf-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Arvind Kothamangala	Arvind 16/09/2026
Dev Lead	Yashas K C	Yashas 16/09/2026
Project Owner/Guide	B Anusha	Anusha 16/09/2026